



AQ8.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Which of these sets form an orthonormal basis of R^3 with respect to the dot product as the inner product in R^3 ?

☐

$\{(1, 0, 0), (0, 1, 0)\}$

☐

$\{(2, 0, 0), (0, 2, 0), (0, 0, 2)\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$

☐

$\{(2, 1, -1), (-1, -1, 1), (1, 2, -2)\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

☐

$\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$



$\sqrt{1, 6}$
 $\sqrt{1, 6}$
 $\sqrt{2), (11}$
 $\sqrt{-3, 11}$
 $\sqrt{1, 11}$
 $\sqrt{1), (66}$
 $\sqrt{1, 66}$
 $\sqrt{7, 66}$
 $\sqrt{-4)\}$
 $\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\} \{(-2$
 $\sqrt{1, 0, 2}$
 $\sqrt{-1), (0, 1, 0), (2$
 $\sqrt{1, 0, 2}$
 $\sqrt{1)\}$

1 point

Consider two orthogonal vectors a, b in $\mathbb{R}^2$. If $a + b$ and $a - b$ are orthogonal, then choose the correct option.

☐

$\|a\| = \|b\| = 1$

☐

$\|a\| = \|b\|$

☐

$\|a\| = 2\|b\|$

☐

$2\|a\| = \|b\|$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\|a\| = \|b\|$

1 point

Consider $a = (1, 1)$, $b = (1, -1)$. Let $V = \text{Span}\{a, b\}$. Choose the correct options by considering the standard inner product (dot product).

☐

Vectors a, b form an orthogonal basis for V .

☐

Vectors a, b form an orthonormal basis for V .

☐

There exist scalar multiples of a, b, c which form an orthonormal basis

☐

There do not exist scalar multiples of a, b, c which form an orthonormal basis

No, the answer is incorrect.

Score: 0

Accepted Answers:

Vectors a, b, c form an orthogonal basis for V .

There exist scalar multiples of a, b, c which form an orthonormal basis

Level 2

1 point

Choose the set of correct statements.

☐

The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is ± 1 .

☐

The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is 0.

☐

The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is ± 1 .

☐

The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is ± 1 .

The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is ± 1 .

1 point

Consider a system of linear equations:

$$2x_1 + 2x_2 + 7x_3 = b_1 \quad 2x_1 + 2x_2 + 7x_3 = b_1$$

$$2x_1 + x_2 - 10x_3 = b_2 \quad 2x_1 + x_2 - 10x_3 = b_2$$

$$3x_1 - 2x_2 + 2x_3 = b_3 \quad 3x_1 - 2x_2 + 2x_3 = b_3.$$

Let A be the coefficient matrix of the given system of linear equations. Let a matrix B contain the column vectors of A , which are normalized by their respective norms, as its columns (i.e. first column vector of A normalized by its norm is the first column of B). Which of the following statements are true ?

☐

The determinant of BB^T is 11.

☐

BB^T is an identity matrix.

☐

BB^T is a scalar matrix.

☐

BB^T is a diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of BB^T is 11.

BB^T is an identity matrix.

BB^T BBT is a scalar matrix.

BB^T BBT is a diagonal matrix.

1 point

Choose the correct option(s).

- ☐ The vectors in an orthonormal set are linearly independent.
- ☐ A set of linearly dependent vectors can be orthonormal.
- ☐ A set of linearly independent vectors is always orthonormal.
- ☐ A set of linearly independent vectors is always orthogonal but not orthonormal.
- ☐ The vectors in an orthogonal set are linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The vectors in an orthonormal set are linearly independent.

The vectors in an orthogonal set are linearly independent.

Consider the inner product $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$ $\langle u, v \rangle = 3u_1 v_1 + 2u_2 v_2$ where

$u = (u_1, u_2)$, $v = (v_1, v_2)$ $u = (u_1, u_2)$, $v = (v_1, v_2)$ are vectors in R^2 . Answer questions 7 and 8 based on this information.

1 point

Which of the following is an orthonormal basis with respect to the inner product defined above?

☐

$\{(2, 3), (-4, 4)\}$

☐

$\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$

☐

$\{(2, 3), (-4, 4)\}$

☐

$\{(2, 3), (-4, 4)\}$

☐

$\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{32}}(-4, 4)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{13}}(-3, 2)\}$

☐

$\{(2, 3), (-3, 2)\}$

☐

$\{(2, 3), (-3, 2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$

☐

$\{(2, 3), (-4, 4)\}$

☐

$$1(-4, 4)\}$$

1 point

Use the orthonormal basis $\{u, v\}$ obtained in question 7 with respect to the defined inner product.

Express the vector $(4, 0)$ as a linear combination of the basis vectors u and v , as

$(4, 0) = c_1 u + c_2 v$. Which of the following gives the coefficients of the linear combination?

☐

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$-48$$

☐

$$c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}} c_1 = 13$$

☐

$$24, c_2 = 32$$

☐

$$-48$$

☐

$$c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}} c_1 = 13$$

☐

$$8, c_2 = 32$$

☐

$$-16$$

☐

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$48$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}} c_1 = 30$$

☐

$$24, c_2 = 80$$

☐

$$-48$$

[Check Answers](#)

Your score is: 0/8